GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of seed mass per plot (esy4 = 4qm plots)

MB

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Contents

> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
```

Packages

```
## Warning: Paket 'ggplot2' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'purrr' wurde unter R Version 4.5.3 erstellt
## Warning: Paket 'dplyr' wurde unter R Version 4.5.3 erstellt
```

```
library(ggbeeswarm)
library(patchwork)
library(DHARMa)
library(emmeans)
```

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
      ),
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
  mutate(
    esy4 = fct_relevel(esy4, "R22", "R1A"),
    eco.id = factor(eco.id)
    ) %>%
  rename(y = cwm.abu.seedmass)
```

Load data

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##       upper        mean       lower
## 0.001621297 0.001509416 0.001397534
```

```
median(sites$y)
```

```
## [1] 0.00126
```

```
sd(sites$y)
```

```
## [1] 0.0009697053
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##       5%      95%
## 0.000375 0.003310
```

```r
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
##   <fct>   <int>
## 1 654       101
## 2 664        91
## 3 686        99
```

```r
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type     n
##   <ord>     <int>
## 1 positive     45
## 2 restored    221
## 3 negative     25
```

```r
sites %>% count(esy4)
```

```
## # A tibble: 2 x 2
##   esy4      n
##   <fct> <int>
## 1 R22     210
## 2 R1A      81
```

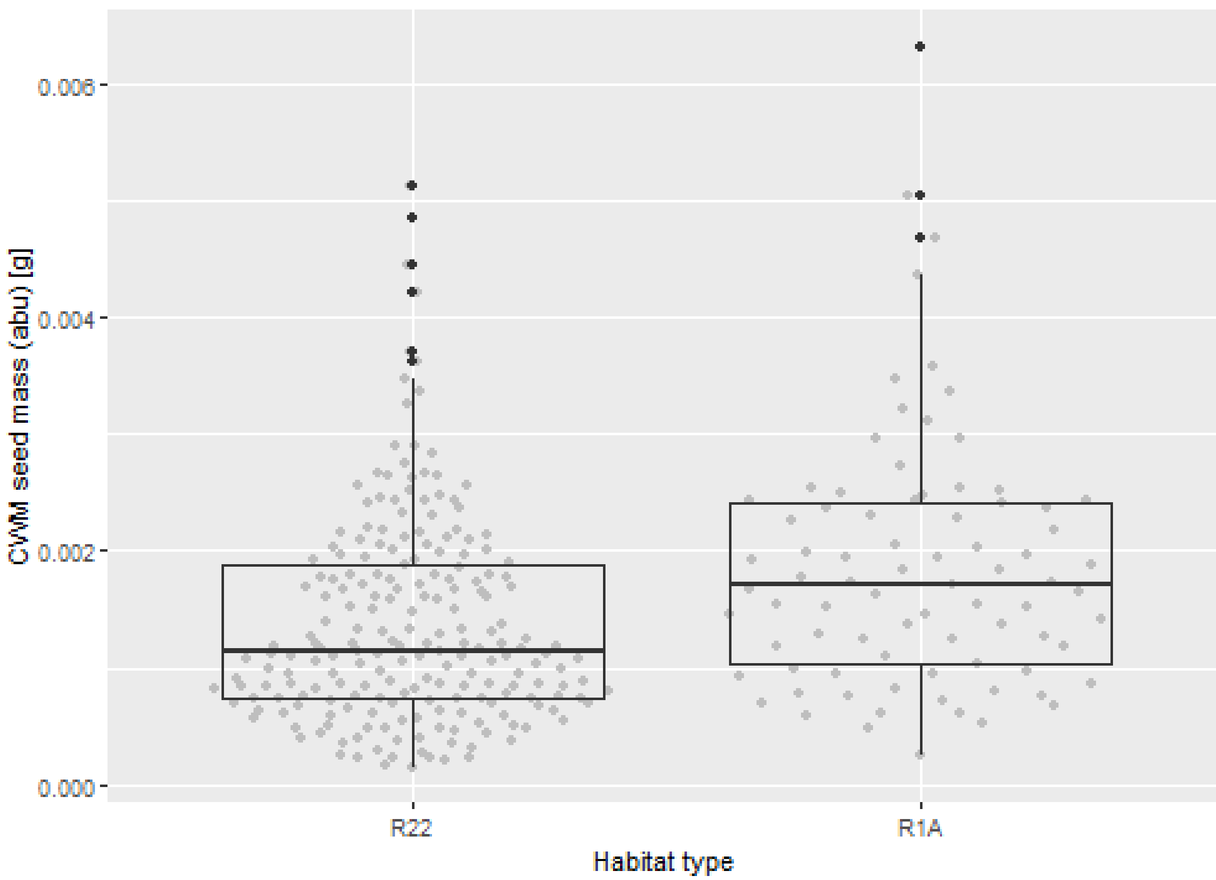
```r
sites %>% count(esy4, eco.id)
```

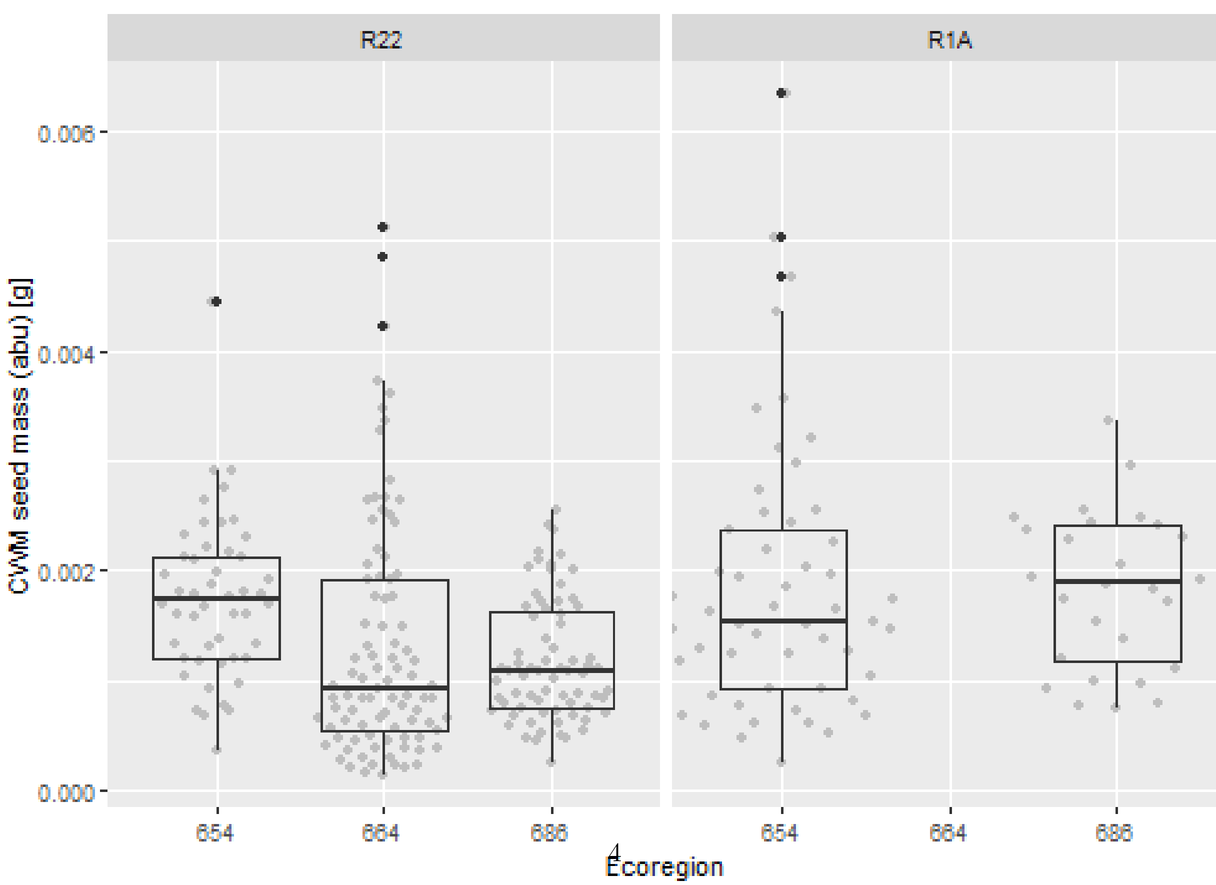
```
## # A tibble: 5 x 3
##   esy4  eco.id      n
##   <fct> <fct>   <int>
## 1 R22   654        48
## 2 R22   664        91
## 3 R22   686        71
## 4 R1A   654        53
## 5 R1A   686        28
```

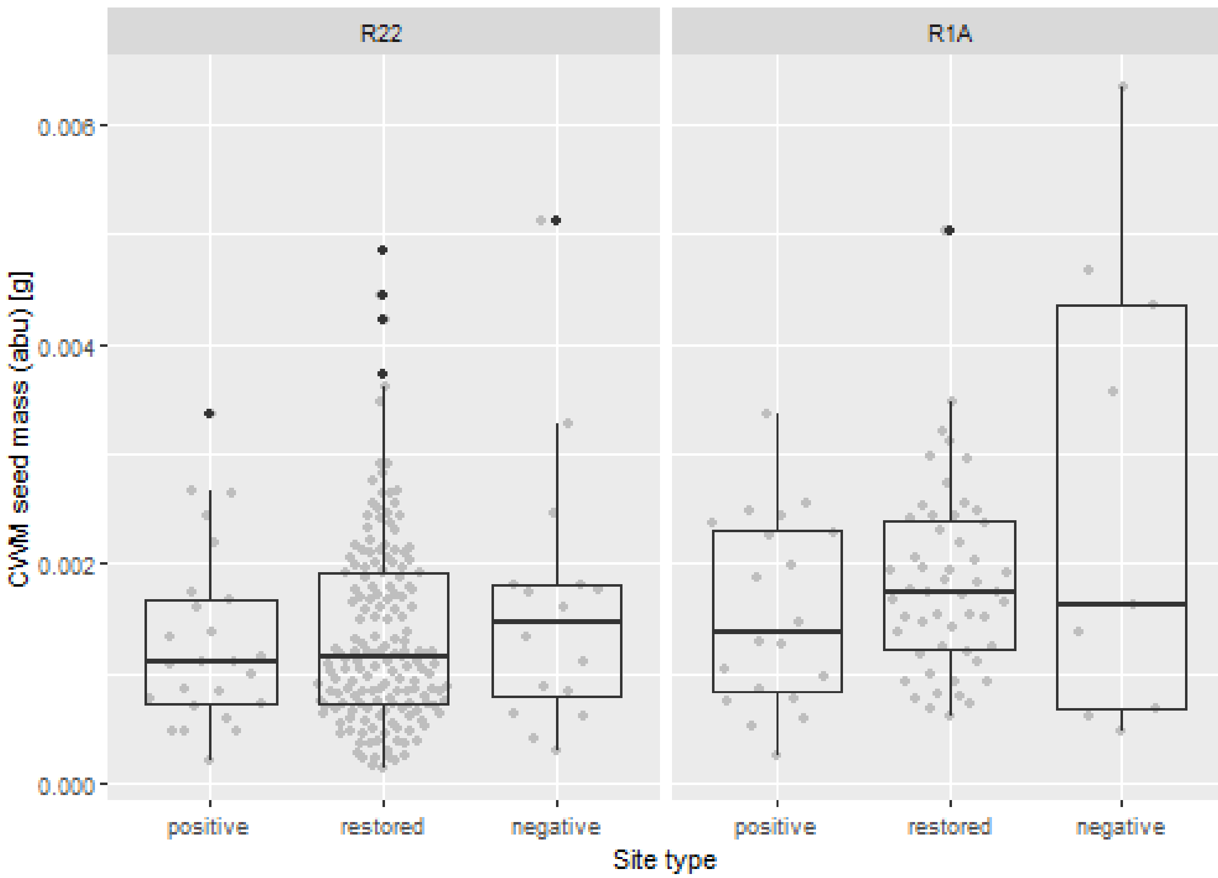
```r
sites %>% count(esy4, site.type)
```

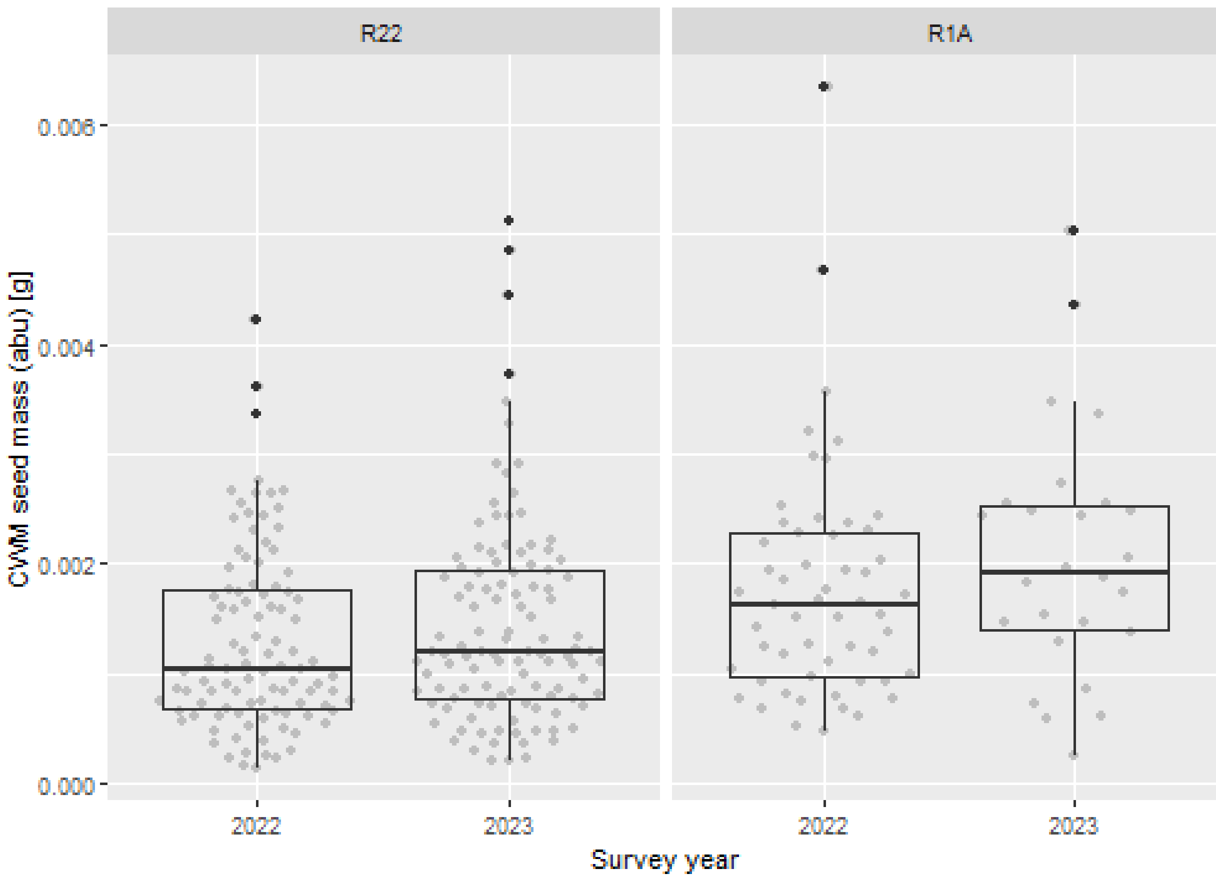
```
## # A tibble: 6 x 3
##   esy4  site.type     n
##   <fct> <ord>     <int>
## 1 R22   positive     25
## 2 R22   restored    169
## 3 R22   negative     16
## 4 R1A   positive     20
## 5 R1A   restored     52
## 6 R1A   negative      9
```

Graphs of raw data (Step 2, 6, 7)

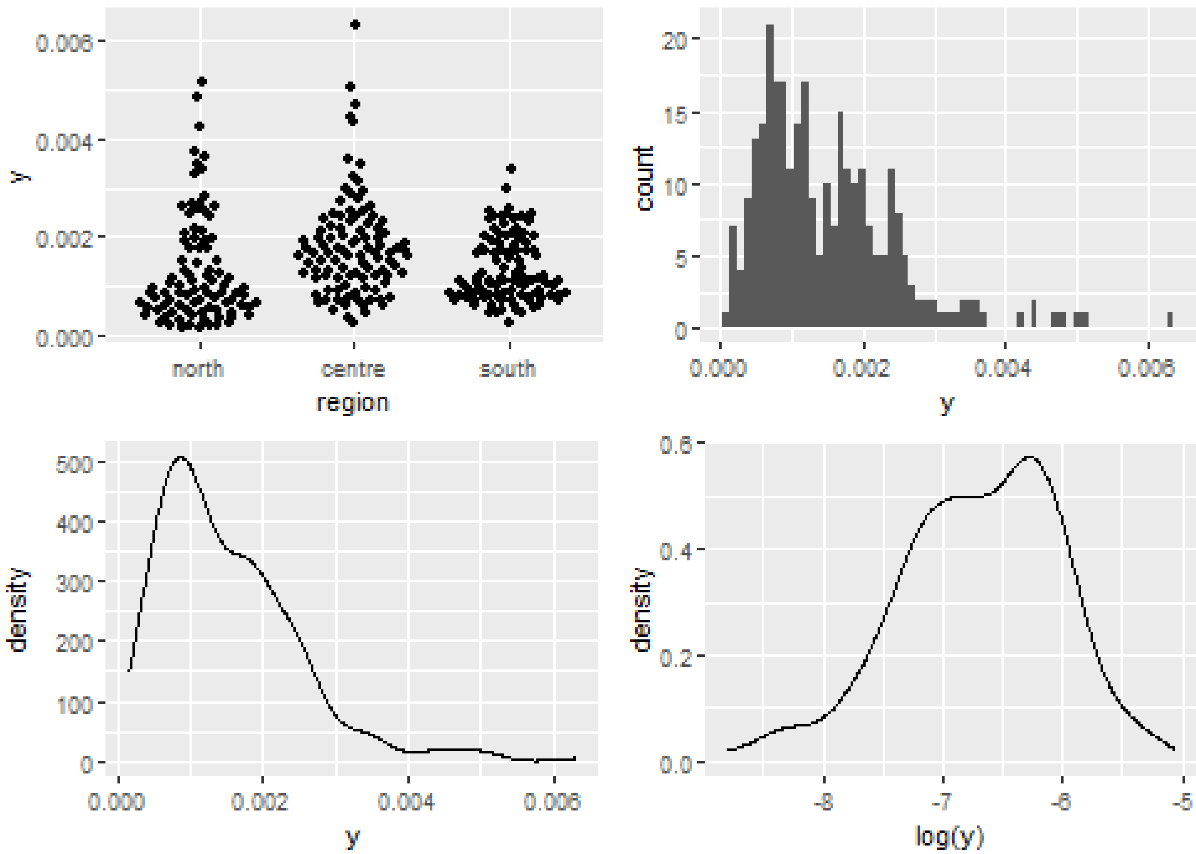





R22
R1A
0.006
0.004
0.002
0.000
CWM seed mass (abu) [g]
positive
restored
negative
positive
restored
negative
Site type


R22
R1A
0.006
0.004
0.002
0.000
CWM seed mass (abu) [g]
2022
2023
2022
2023
Survey year

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_seedmass_cwm_1.Rdata"))
load(file = here("outputs", "models", "model_seedmass_cwm_2.Rdata"))
m_1 <- m1
m_2 <- m2
```

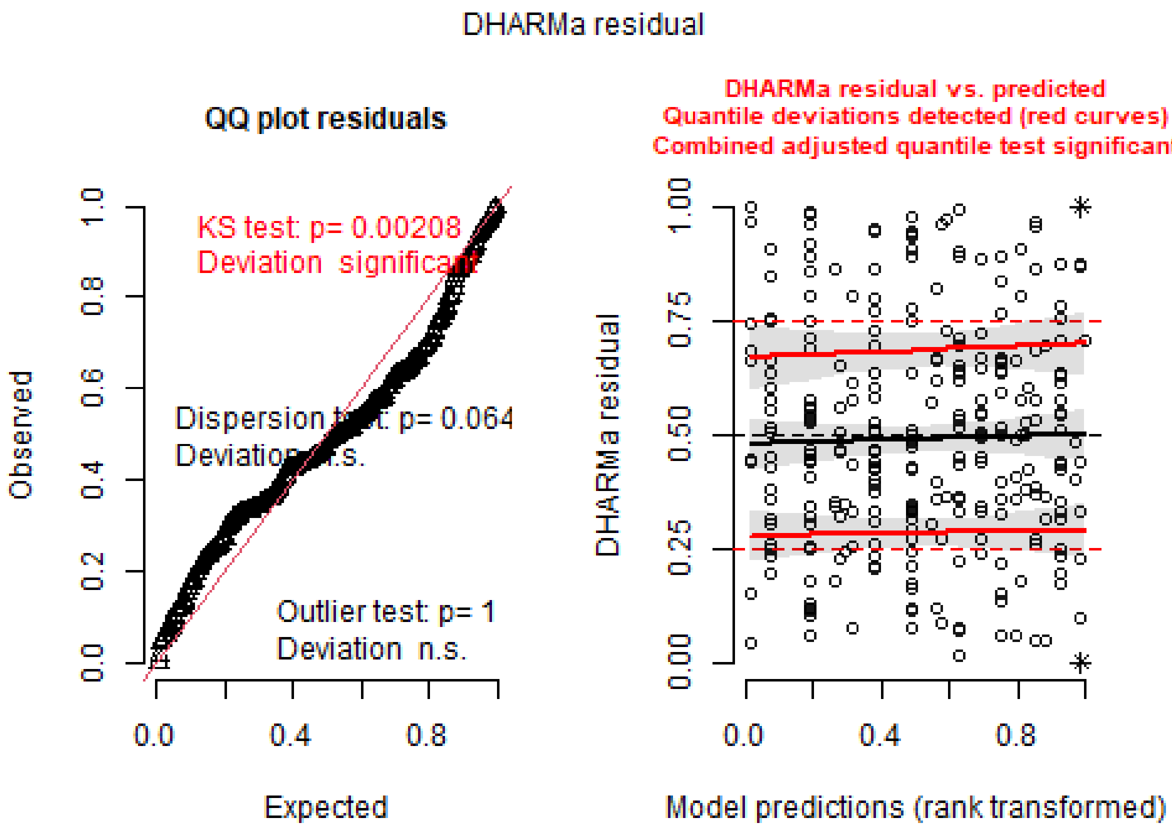
```
m_1@call
## lmer(formula = sqrt(y) ~ esy4 * (site.type + eco.id) + obs.year +
```

```
##     (1 | id.site), data = sites, REML = FALSE)
m_2@call
## lmer(formula = sqrt(y) ~ (esy4 + site.type + eco.id + obs.year)^2 +
##     (1 | id.site), data = sites, REML = FALSE)
```

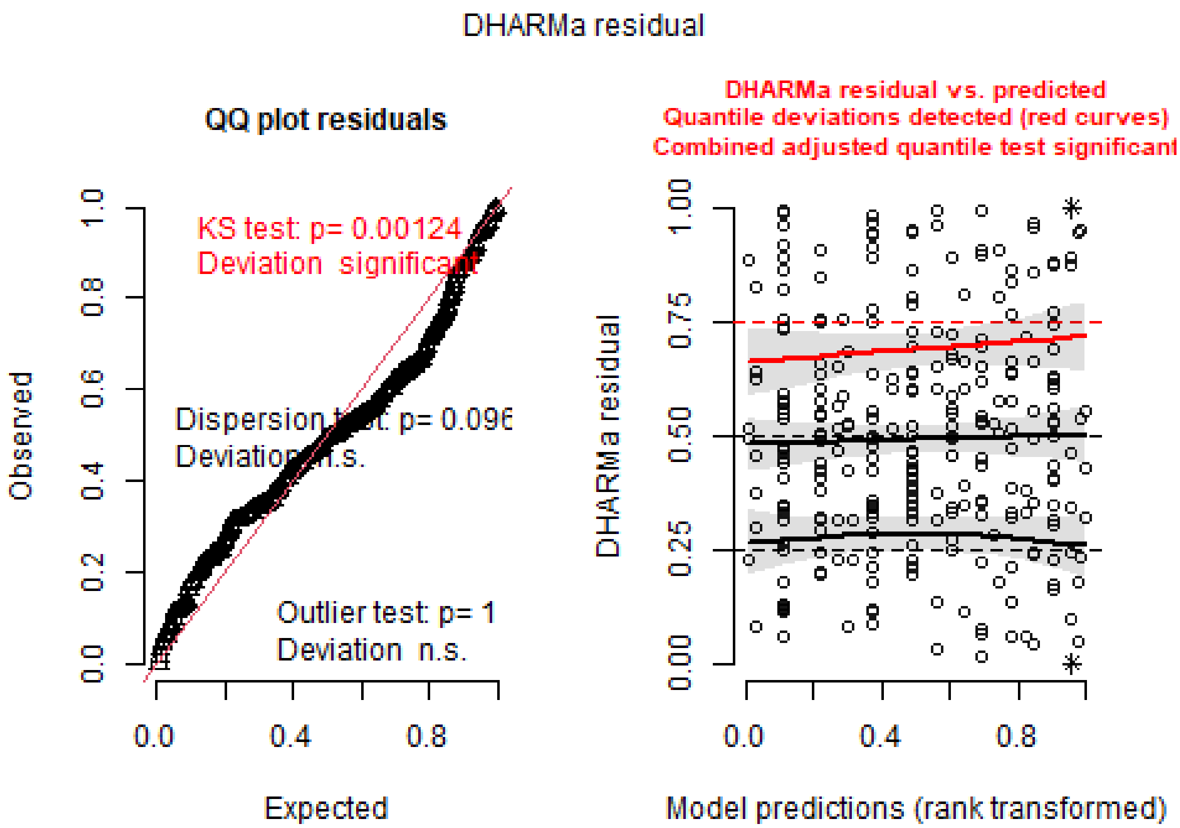
Model check

DHARMa

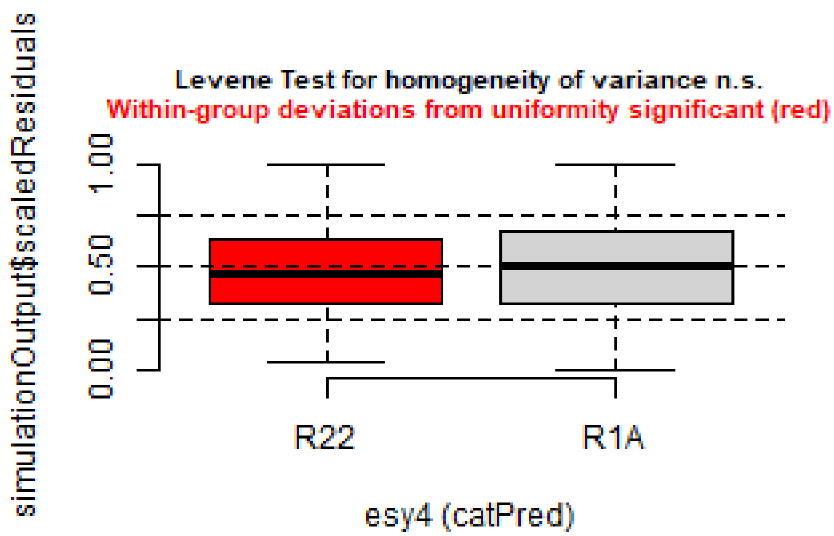
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



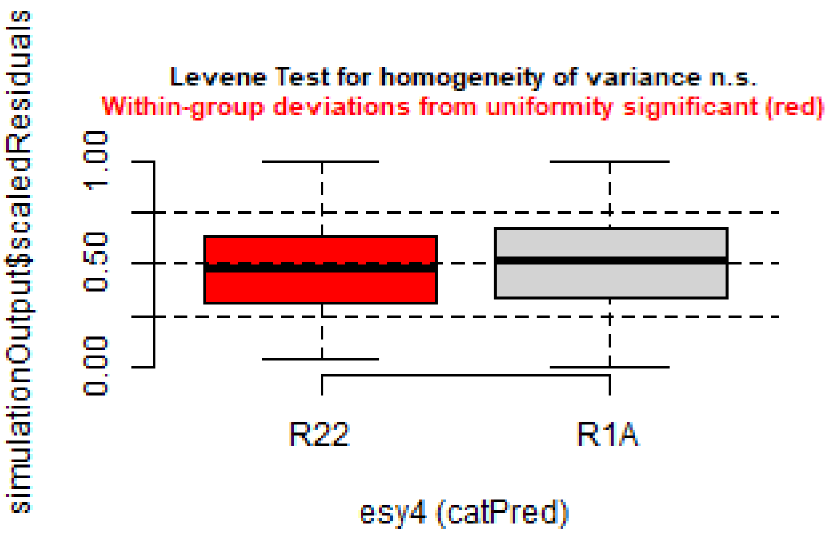

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



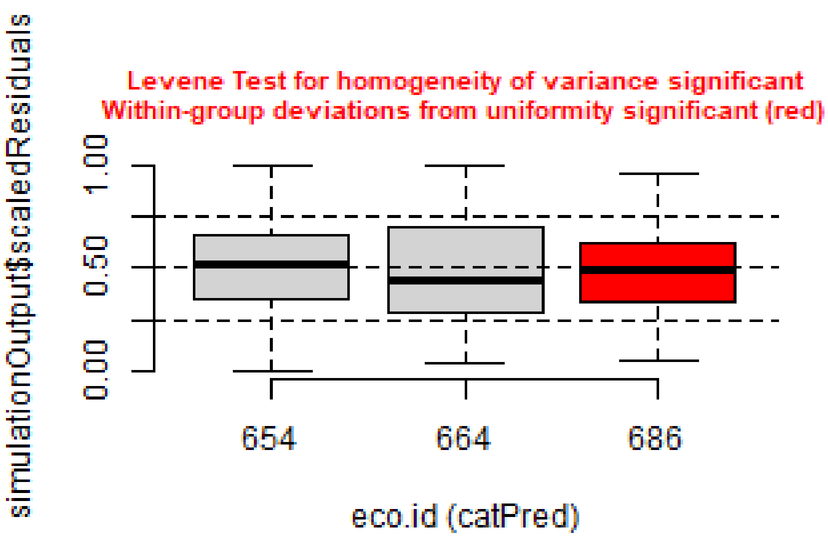

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



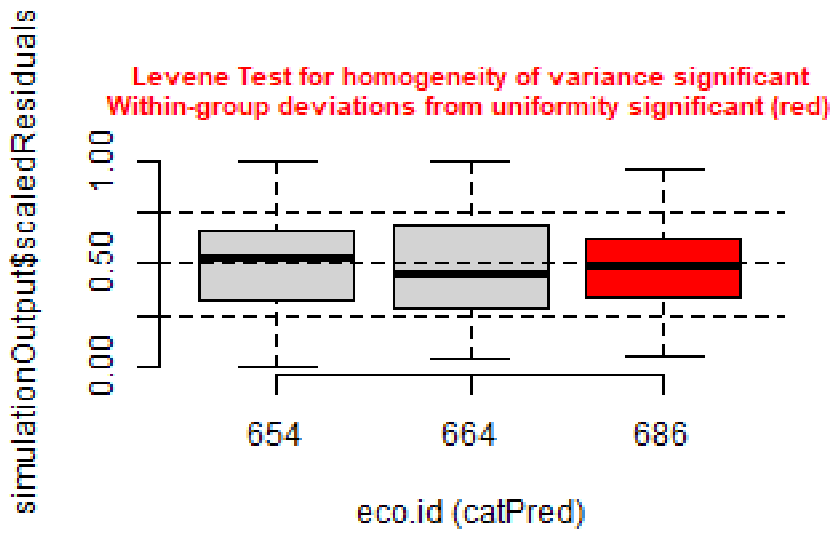

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



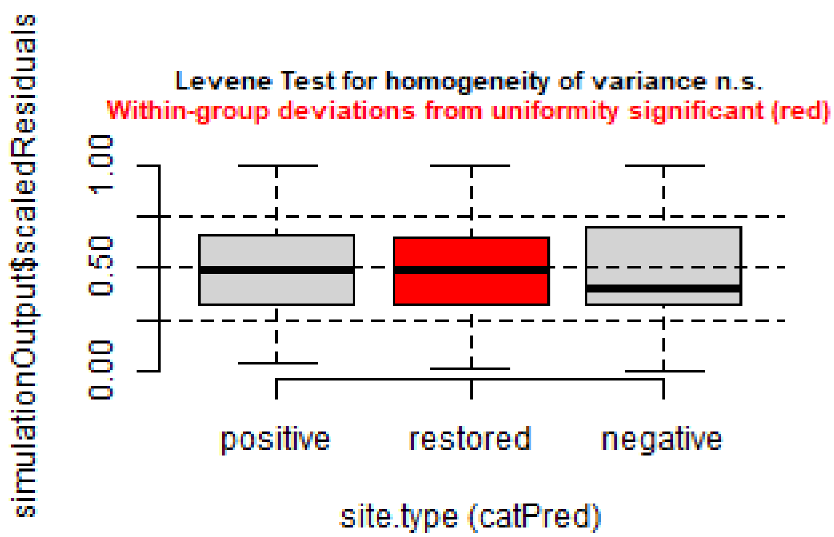

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



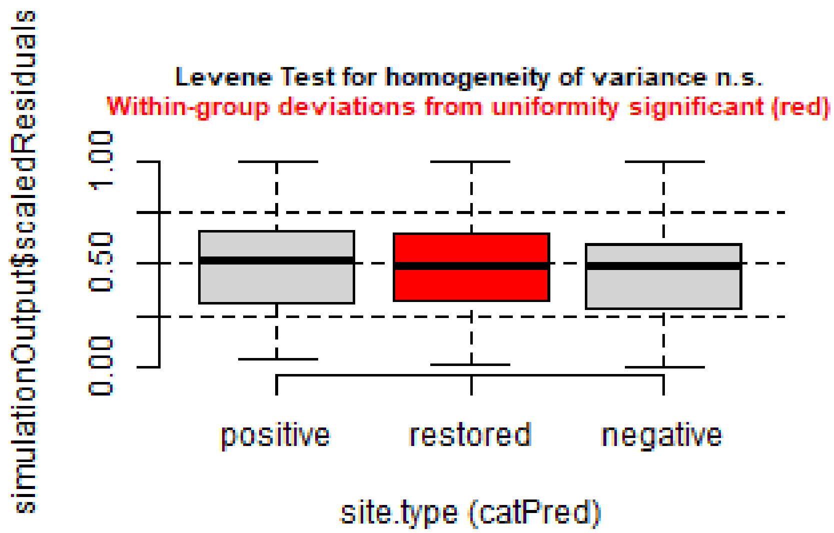

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



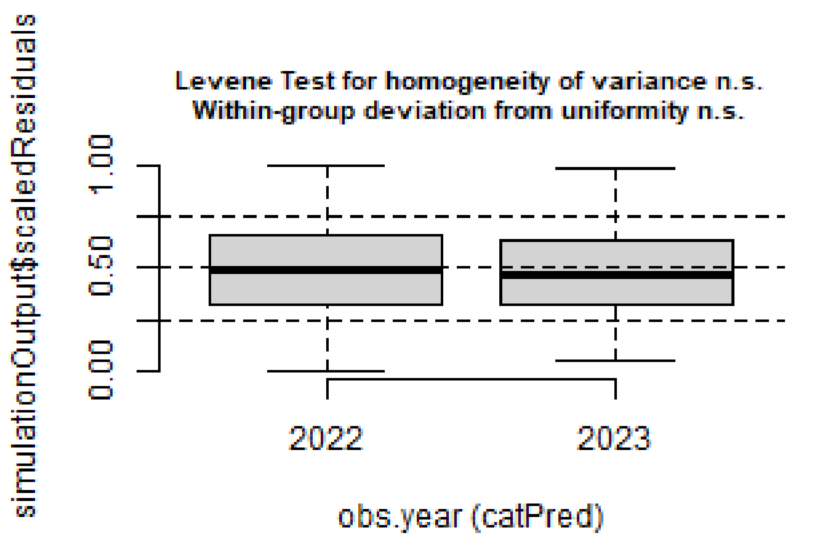

```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



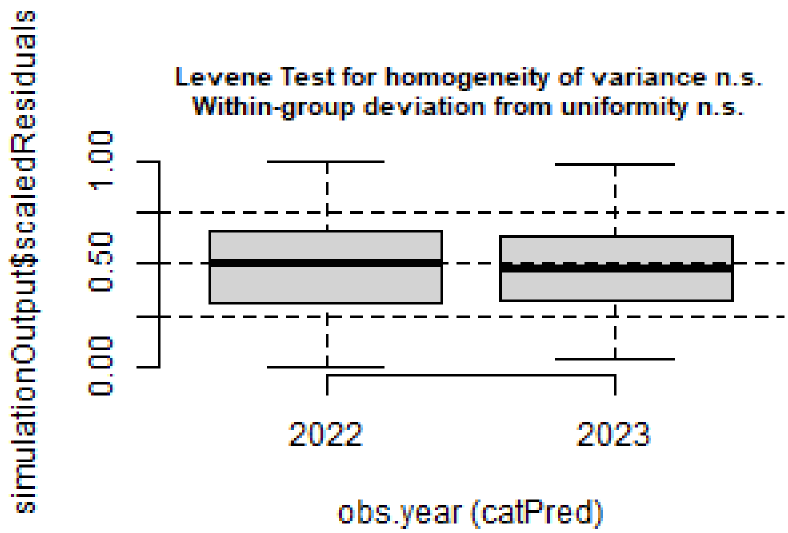

```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



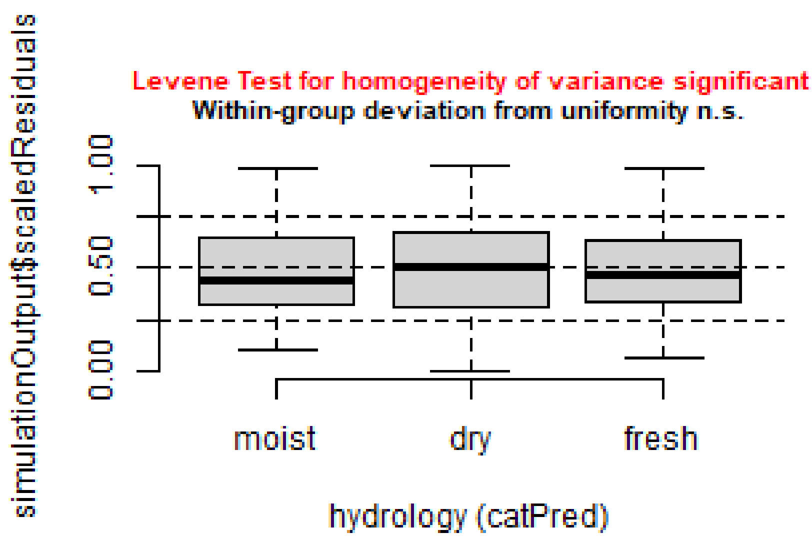

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



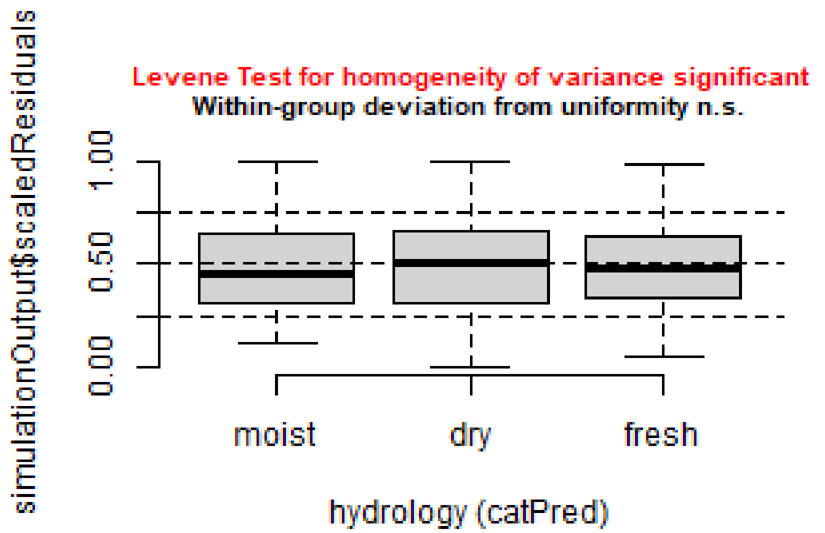

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



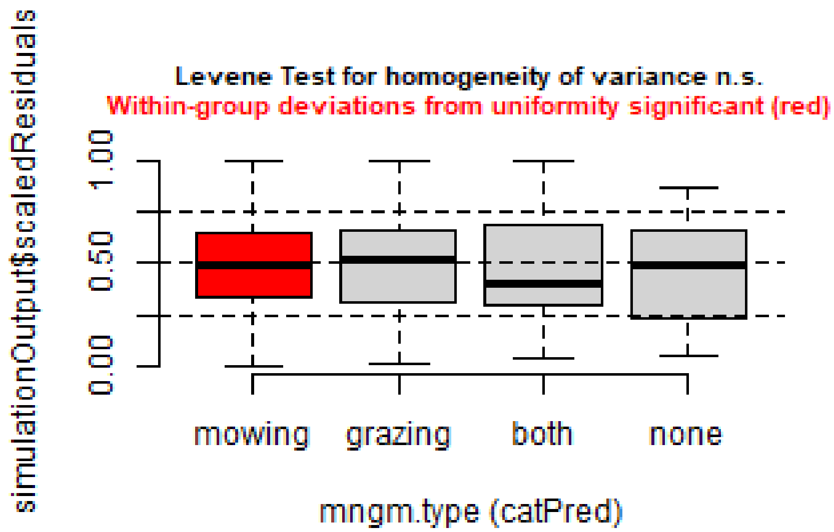

```r
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



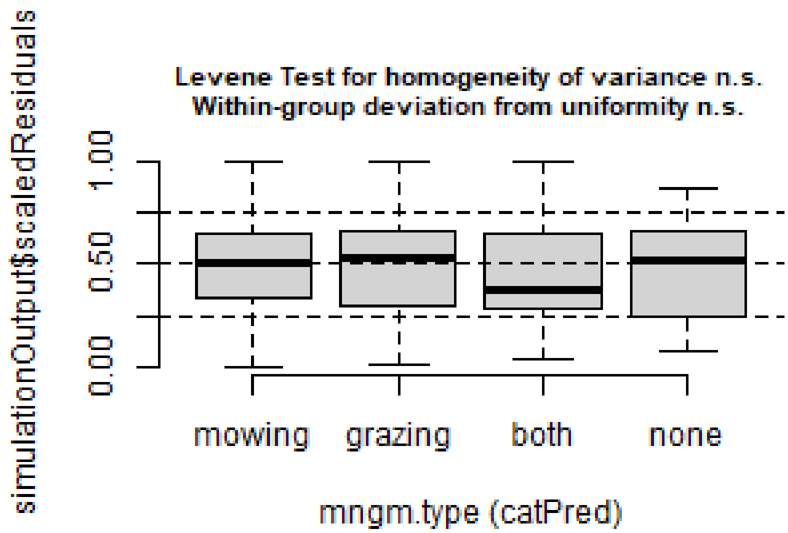

```r
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```




```r
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```




```r
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
## Registered S3 method overwritten by 'car':
##   method          from
##   na.action.merMod lme4

##                    GVIF Df GVIF^(1/(2*Df))
## esy4           3.187659  1        1.785402
## site.type      1.793242  2        1.157203
## eco.id         1.622014  2        1.128531
## obs.year       1.041591  1        1.020584
## esy4:site.type 3.254741  2        1.343164
## esy4:eco.id    2.280958  1        1.510284
```

```
car::vif(m_2)
```

```
##                         GVIF Df GVIF^(1/(2*Df))
## esy4                5.484399  1        2.341879
## site.type          50.609397  2        2.667214
## eco.id             18.938919  2        2.086118
## obs.year            5.906422  1        2.430313
## esy4:site.type      8.163884  2        1.690341
## esy4:eco.id         2.584404  1        1.607608
## esy4:obs.year       3.278605  1        1.810692
## site.type:eco.id   67.568066  4        1.693237
## site.type:obs.year  9.506261  2        1.755911
## eco.id:obs.year    11.152354  2        1.827434
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##            R2m       R2c
## [1,] 0.1326057 0.7336008
MuMIn::r.squaredGLMM(m_2)
##            R2m      R2c
## [1,] 0.1888485 0.729805
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df      AICc
## m_1 12 -1892.558
## m_2 21 -1885.331
```

Predicted values

Summary table

```
car::Anova(m_1, type = 3)
```

```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: sqrt(y)
##                   Chisq Df Pr(>Chisq)
## (Intercept)    230.8539  1  < 2.2e-16 ***
## esy4             0.3010  1   0.583234
## site.type        0.5747  2   0.750243
## eco.id          10.4419  2   0.005402 **
## obs.year         3.3092  1   0.068891 .
## esy4:site.type   1.9203  2   0.382830
## esy4:eco.id      4.4416  1   0.035073 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(m_1)
```

```
## Linear mixed model fit by maximum likelihood  ['lmerMod']
## Formula: sqrt(y) ~ esy4 * (site.type + eco.id) + obs.year + (1 | id.site)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##   -1893.7   -1849.6     958.8   -1917.7       279
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.2799 -0.4744 -0.0583  0.4011  3.0144
##
```

```
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  id.site  (Intercept) 8.717e-05 0.009336
##  Residual             3.864e-05 0.006216
## Number of obs: 291, groups:  id.site, 119
##
## Fixed effects:
##                       Estimate Std. Error t value
## (Intercept)          0.0397857  0.0026185  15.194
## esy4R1A              0.0018396  0.0033528   0.549
## site.type.L          0.0021493  0.0030892   0.696
## site.type.Q         -0.0003436  0.0020223  -0.170
## eco.id664           -0.0072867  0.0027546  -2.645
## eco.id686           -0.0082658  0.0026951  -3.067
## obs.year2023         0.0035132  0.0019313   1.819
## esy4R1A:site.type.L  0.0069769  0.0056336   1.238
## esy4R1A:site.type.Q  0.0042527  0.0037653   1.129
## esy4R1A:eco.id686    0.0086946  0.0041255   2.108
##
## Correlation of Fixed Effects:
##             (Intr) es4R1A st.t.L st.t.Q ec.664 ec.686 o.2023 e4R1A:..L
## esy4R1A     -0.606
## site.type.L  0.009 -0.081
## site.type.Q  0.440 -0.307  0.181
## eco.id664   -0.661  0.438  0.112 -0.050
## eco.id686   -0.632  0.436  0.049 -0.020  0.596
## obs.yer2023 -0.422  0.078  0.120 -0.025  0.076  0.008
## esy4R1A:..L -0.101  0.334 -0.481 -0.155 -0.009  0.014  0.015
## esy4R1A:..Q -0.220  0.500 -0.153 -0.463  0.034  0.028  0.004  0.468
## es4R1A:.686  0.338 -0.503 -0.048  0.009 -0.329 -0.499  0.018  0.166
##             e4R1A:..Q
## esy4R1A
## site.type.L
## site.type.Q
## eco.id664
## eco.id686
## obs.yer2023
## esy4R1A:..L
## esy4R1A:..Q
## es4R1A:.686  0.148
## fit warnings:
## fixed-effect model matrix is rank deficient so dropping 1 column / coefficient
```

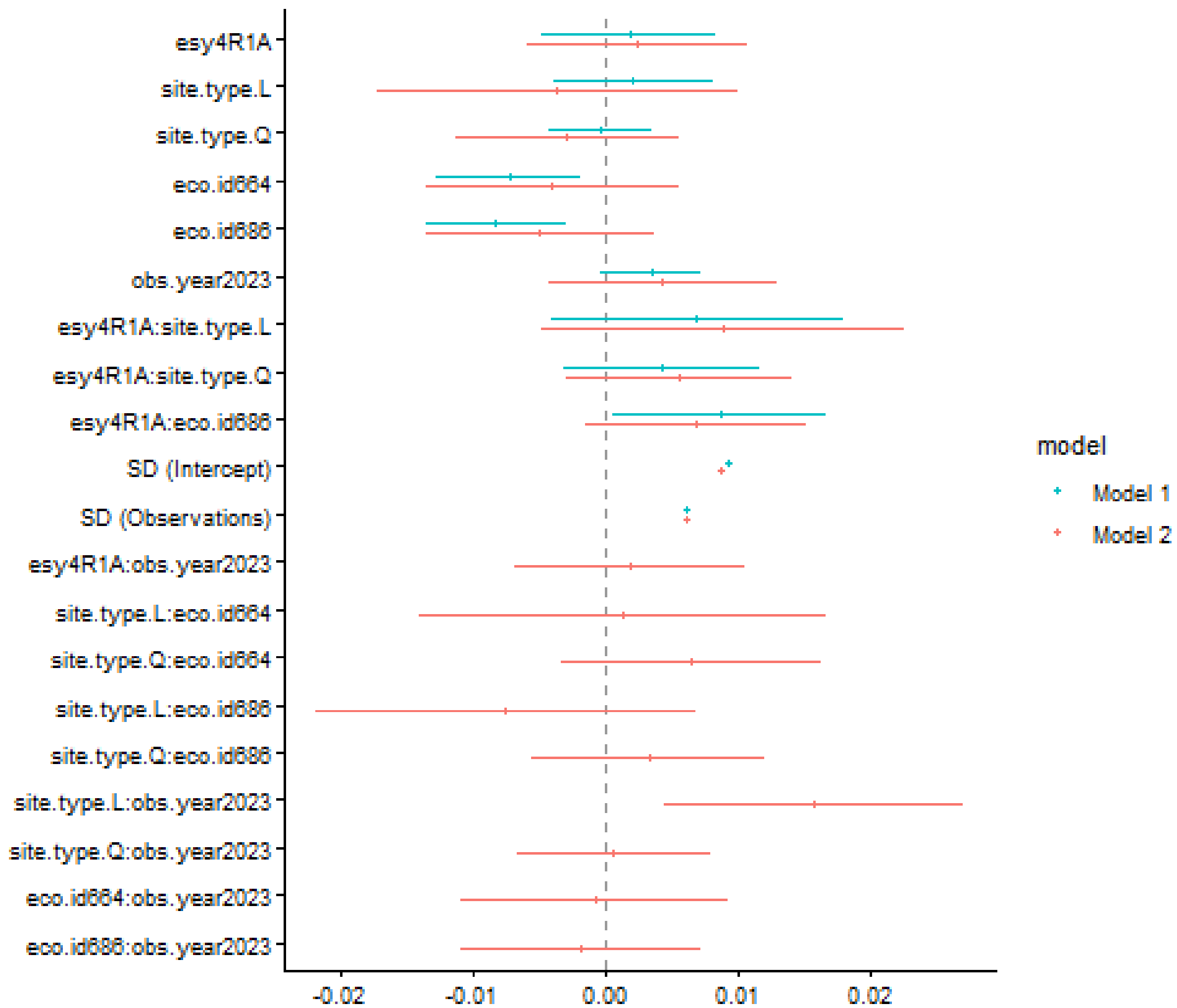
Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.

## Warning: Using the `size` aesthetic with geom_segment was deprecated in ggplot2 3.4.0.
## i Please use the `linewidth` aesthetic instead.
## i The deprecated feature was likely used in the dotwhisker package.
##   Please report the issue at <https://github.com/fsolt/dotwhisker/issues>.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



Effect sizes: Habiat type x Region

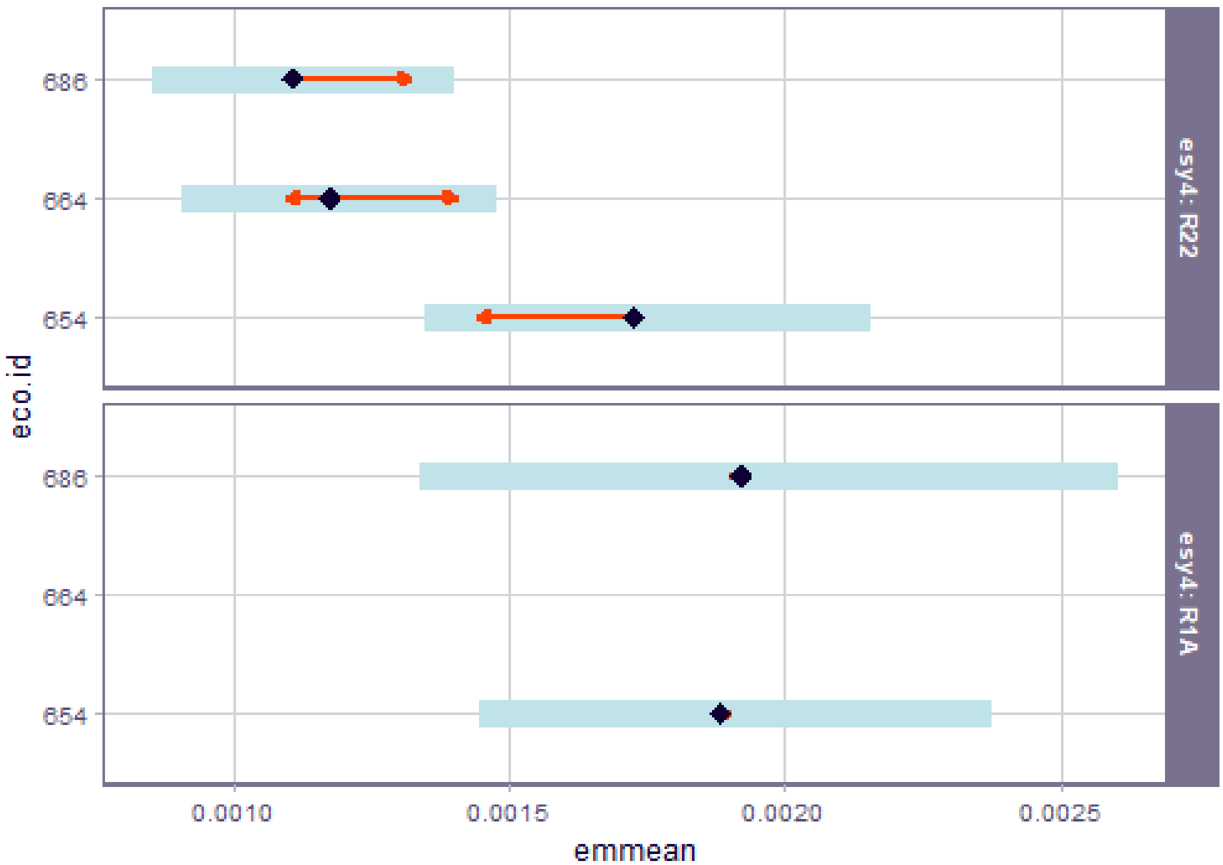
```
emm <- emmeans(
  m_1,
  revpairwise ~ eco.id | esy4,
  type = "response"
  )
plot(emm, comparison = TRUE)
```

```
## Warning: Comparison discrepancy in group "R1A", eco.id654 - eco.id686:
##     Target overlap = 0.9422, overlap on graph = -0.0195

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_segment()`).

## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
emm
```

```
## $emmeans
## esy4 = R22:
##  eco.id response       SE  df lower.CL upper.CL
##  654      0.00173 0.000205 164 0.001345  0.00215
##  664      0.00117 0.000145 130 0.000905  0.00148
##  686      0.00111 0.000139 156 0.000850  0.00140
##
## esy4 = R1A:
##  eco.id response       SE  df lower.CL upper.CL
##  654      0.00188 0.000234 141 0.001447  0.00237
##  664       nonEst       NA  NA       NA       NA
##  686      0.00192 0.000321 186 0.001338  0.00260
##
## Results are averaged over the levels of: site.type, obs.year
```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
## Intervals are back-transformed from the sqrt scale
##
## $contrasts
## esy4 = R22:
##  contrast              estimate      SE  df t.ratio p.value
##  eco.id664 - eco.id654 -0.007287 0.00286 146  -2.546  0.0318
##  eco.id686 - eco.id654 -0.008266 0.00280 159  -2.955  0.0100
##  eco.id686 - eco.id664 -0.000979 0.00255 139  -0.385  0.9218
##
## esy4 = R1A:
##  contrast              estimate      SE  df t.ratio p.value
##  eco.id664 - eco.id654    nonEst      NA  NA      NA      NA
##  eco.id686 - eco.id654  0.000429 0.00377 201   0.114  0.9094
##  eco.id686 - eco.id664    nonEst      NA  NA      NA      NA
##
## Results are averaged over the levels of: site.type, obs.year
## Note: contrasts are still on the sqrt scale. Consider using
##       regrid() if you want contrasts of back-transformed estimates.
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for varying family sizes
```

Session info

```
## R version 4.5.2 (2025-10-31 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_2.0.3    DHARMa_0.5.0     patchwork_1.3.2  ggbeeswarm_0.7.3
##  [5] lubridate_1.9.5  forcats_1.0.1    stringr_1.6.0    dplyr_1.2.1
##  [9] purrr_1.2.2      readr_2.2.0      tidyr_1.3.2      tibble_3.3.1
## [13] ggplot2_4.0.3    tidyverse_2.0.0  here_1.0.2
##
## loaded via a namespace (and not attached):
##  [1] Rdpack_2.6.6          gridExtra_2.3         sandwich_3.1-1
##  [4] rlang_1.2.0           magrittr_2.0.5        multcomp_1.4-30
##  [7] otel_0.2.0            compiler_4.5.2        mgcv_1.9-3
```

```
## [10] vctrs_0.7.3             pkgconfig_2.0.3        crayon_1.5.3
## [13] fastmap_1.2.0           backports_1.5.1        labeling_0.4.3
## [16] utf8_1.2.6              ggstance_0.3.7         promises_1.5.0
## [19] rmarkdown_2.31          tzdb_0.5.0             nloptr_2.2.1
## [22] bit_4.6.0               xfun_0.58              later_1.4.8
## [25] broom_1.0.13            parallel_4.5.2         R6_2.6.1
## [28] gap.datasets_0.0.6      stringi_1.8.7          qgam_2.0.0
## [31] RColorBrewer_1.1-3      car_3.1-5              boot_1.3-32
## [34] estimability_1.5.1      Rcpp_1.1.1-1.1         iterators_1.0.14
## [37] knitr_1.51              zoo_1.8-15             parameters_0.29.1
## [40] httpuv_1.6.17           Matrix_1.7-4           splines_4.5.2
## [43] timechange_0.4.0        tidyselect_1.2.1       rstudioapi_0.18.0
## [46] abind_1.4-8             yaml_2.3.12            MuMIn_1.48.19
## [49] doParallel_1.0.17       codetools_0.2-20       lattice_0.22-7
## [52] plyr_1.8.9              bayestestR_0.18.1      shiny_1.13.0
## [55] withr_3.0.2             S7_0.2.2               evaluate_1.0.5
## [58] marginaleffects_0.32.0 survival_3.8-3         pillar_1.11.1
## [61] gap_1.15.2              carData_3.0-6          foreach_1.5.2
## [64] stats4_4.5.2            insight_1.5.1          reformulas_0.4.4
## [67] generics_0.1.4          vroom_1.7.1            rprojroot_2.1.1
## [70] hms_1.1.4               scales_1.4.0           minqa_1.2.8
## [73] xtable_1.8-8            glue_1.8.1             tools_4.5.2
## [76] data.table_1.18.4       lme4_2.0-1             mvtnorm_1.4-0
## [79] grid_4.5.2              rbibutils_2.4.1        datawizard_1.3.1
## [82] nlme_3.1-168            Rmisc_1.5.1            performance_0.17.0
## [85] beeswarm_0.4.0          vipor_0.4.7            Formula_1.2-5
## [88] cli_3.6.6               gtable_0.3.6           digest_0.6.39
## [91] pbkrtest_0.5.5          TH.data_1.1-5          farver_2.1.2
## [94] htmltools_0.5.9         lifecycle_1.0.5        mime_0.13
## [97] bit64_4.8.2             dotwhisker_0.8.4       MASS_7.3-65
```